

# A 1:10 Autonomous Car for the Team

Budget breakdown under \$2,000

Formula Electric at Berkeley, Autonomous Team

September 2026

## Summary

We ask for **\$1,750** to build one 1:10-scale autonomous car and a lab track, with a minimum configuration at **\$1,100** if the budget is tighter. The car is the RoboRacer-standard Traxxas Slash with a 2D lidar, a small computer and a motor controller that reports wheel speed. It is what new members learn real robotics on, what the perception and control work is tested on between FSAE seasons, and the first place our code meets real dynamics. The team's simulator already exists and is in use; this document is about the hardware, area by area, with the costs, the choices inside each area, and the realities that come with them.

| Area                | Recommended  | Minimum      |
|---------------------|--------------|--------------|
| Chassis and drive   | 500          | 330          |
| Motor controller    | 110          | 110          |
| Compute             | 250          | 130          |
| Lidar               | 320          | 100          |
| Camera              | 0            | 0            |
| Power               | 230          | 200          |
| Mounting and wiring | 80           | 60           |
| Track and cones     | 45           | 15           |
| Spares and tools    | 60           | 40           |
| Subtotal            | 1,595        | 985          |
| Contingency (10 %)  | 160          | 100          |
| <b>Total</b>        | <b>1,755</b> | <b>1,085</b> |

USD, September 2026 list prices, rounded. Verify at order time; the lidar moves most.

## 1 Where we are

- **Simulator.** The team has a working simulator of this exact car, a ROS 2 devkit, a starter driving stack and a leaderboard. Members already write and race code in it. It needs nothing from this budget. Its limit is that nothing in it is real: no grip change with the floor, no servo backlash, no battery sag, no lidar seeing chair legs.
- **Owned hardware.** An Intel RealSense depth camera, a RoboSense M1 3D lidar, and a motorless Tamiya 1:10 touring chassis. The camera goes on this car. The M1 is a full-size-vehicle sensor (about 0.7 kg, half-metre blind zone) and does not belong on a 1:10 car; it waits for a larger platform. The Tamiya has no motor, controller or sensor deck, does not match the simulator's car, and would cost a semester of mechanical work to save about \$300; it stays a bench mule.

- **Space.** A lab corridor is enough for a 1:10 track. The Alameda test track is for larger vehicles and is not needed here.

## 2 The car, area by area

### 2.1 Chassis and drive

| Option                                                               | USD |             |
|----------------------------------------------------------------------|-----|-------------|
| Traxxas Slash 4x4 VXL (TRA68086-4), brushless motor and ESC included | 500 | recommended |
| Traxxas Slash 2WD VXL (TRA58076-4), brushless                        | 330 | minimum     |

The Slash is the chassis every RoboRacer build and drawing assumes, and the one the simulator models. The 4x4 is the standard: it carries the sensor deck without lifting the front wheels and tolerates novice driving. The 2WD saves \$170 and is fine for lidar-only driving on a flat floor, but it spins the rear under a heavy deck and is not what the drawings fit; buy it only if the total must come under \$1,200. Either comes ready to run with a brushless motor; the stock speed controller is replaced by the item below, and the radio and receiver stay as the manual-override link.

### 2.2 Motor controller

| Option                                   | USD |                           |
|------------------------------------------|-----|---------------------------|
| Flipsky mini F5ESC 6.7 (VESC-compatible) | 110 | recommended               |
| VESC 6 MkVI (reference part)             | 260 | not needed at this budget |

A VESC-type controller is not optional: it is what gives the software wheel speed (odometry), current limits, and a clean serial interface. The reference VESC costs \$150 more for better cooling and support; the Flipsky runs the same firmware and is what most student builds use.

### 2.3 Compute

| Option                                            | USD |             |
|---------------------------------------------------|-----|-------------|
| NVIDIA Jetson Orin Nano developer kit             | 250 | recommended |
| Raspberry Pi 5, 8 GB, active cooler, SD card, PSU | 130 | minimum     |

The Pi runs ROS 2 and a lidar-only driver comfortably and is the cheaper way to a first lap. It cannot run camera perception at useful rates, so the day the team wants cone detection from the RealSense, it needs the Jetson. Buying the Jetson first avoids buying twice; buying the Pi first gets a car on the floor for \$120 less and keeps the Jetson as the upgrade.

### 2.4 Lidar

| Option                                     | USD   |                                   |
|--------------------------------------------|-------|-----------------------------------|
| Slamtec RPLidar A2M12, 12 m, 12 Hz, 0.25°  | 320   | recommended                       |
| Slamtec RPLidar A1M8, 12 m, 5 to 10 Hz, 1° | 100   | minimum                           |
| Hokuyo UST-10LX, 10 m, 40 Hz, 0.25°        | 1,500 | the reference; out of this budget |

The lidar decides how fast the car can safely go. The reference Hokuyo scans at 40 Hz and is what the simulator models; it is also most of a \$3,000 build. The A2M12 at 12 Hz is a sensible car for 2 to 3 m/s and is what the budget allows. The A1M8 is a \$100 sensor that proves the pipeline

and nothing more: 8 kHz sample rate, a degree of angular resolution, noisy beyond 6 m. Spinning lidars are also the part that breaks in a crash; mount it behind the bumper line.

## 2.5 Camera

The team's RealSense goes on the car at no cost. With the lidar it gives the sensor pair for cone detection: the camera classifies cones by colour, the lidar gives range and bearing. This is the perception work that carries most directly to the FSAE car, and it needs the Jetson.

## 2.6 Power

| Item                                                 | USD |                                              |
|------------------------------------------------------|-----|----------------------------------------------|
| Two Traxxas 3S 5000 mAh LiPo packs                   | 170 | one drives, one charges                      |
| Balance charger                                      | 50  |                                              |
| LiPo safety bag                                      | 10  |                                              |
| 5 V / 5 A buck converter, XT60 splitter, inline fuse | 30  | feeds computer and lidar from the drive pack |

Minimum: one pack (\$85 less). The RoboRacer powerboard (\$100) is a cleaner version of the converter and splitter with a safe-shutdown line; not needed to start.

## 2.7 Mounting and wiring

Laser-cut deck plates from the published RoboRacer drawings, cut in a campus makerspace (about \$30 of acrylic), standoffs and fasteners (\$15), a powered USB hub and cables (\$35). Minimum \$60 with 3D-printed brackets instead of a cut deck.

## 2.8 Track and cones

| Item                                                           | USD |                                    |
|----------------------------------------------------------------|-----|------------------------------------|
| Cardboard walls: flattened boxes, 20 cm strips, taped in loops | 0   | recommended to start               |
| Floor tape (start line, lane marks)                            | 15  |                                    |
| FSAE-style cones, 3D-printed at 1:10, 100 pieces               | 15  | about 10 g of filament each        |
| Foam pipe insulation or pool noodles as bumpers on corners     | 15  | optional                           |
| Flexible aluminium duct, 33 cm, 30 m (the league's walls)      | 130 | later, if the track gets daily use |

**Cardboard is the right start.** The lidar sees cardboard as well as duct; a 20 cm wall is well above the scan plane; a wall that folds when hit is kinder to the car than one that does not; and the cost is zero. Its faults are real: it gets flattened, it has to be re-taped every session, and it does not store. Duct is what to buy once the team is running the track several times a week and is tired of rebuilding it. The cones are cheap and worth printing on day one, because the cone track is where the camera work happens.

## 2.9 Spares and tools

Tyres (\$30), pinion and bumper (\$20), zip ties, tape, Velcro (\$10). The team's own tools otherwise. A wall hit at 3 m/s breaks a bumper or a servo saver, not the car; a lidar hit breaks the lidar.

### 3 What the money buys, in order

| When               | Milestone                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------|
| Order + 2 weekends | Car assembled from the RoboRacer build guide, radio drive, all sensors on ROS 2                    |
| + 3 weeks          | Starter stack drives the cardboard track; simulator-to-car differences written down                |
| + 6 weeks          | Members' own stacks on the car; internal time-attack; first cone-detection results from the camera |
| Semester end       | Perception and control lessons carried into the FSAE autonomous work                               |

### 4 The realities

- **One car.** Access has to be scheduled and earned; the simulator's qualification rule is the gate. Expect queueing in the weeks before an internal event.
- **It will be crashed.** A spinning lidar on the nose of a car driven by student code is a matter of when. The contingency and the spares line are for this. A dead A2M12 is \$320; a dead Hokuyo would be \$1,500, which is the other reason it is not in this budget.
- **The cheap parts cap performance.** At 12 Hz the lidar and the Pi limit safe speed to about 2 to 3 m/s; the simulator car goes faster. Members will feel the gap; that is the point, but it should be said up front.
- **Time is the scarce resource, not money.** Two weekends of assembly, a month to a first clean lap, and someone has to own the car, the batteries and the charging procedure. If nobody owns it, it becomes a shelf ornament by week six.
- **Batteries need a procedure.** LiPo bag, supervised charging, no charging unattended, a written checklist before the first charge.
- **The floor matters.** Lab tile and carpet grip differently; the car will be tuned for one floor. A cardboard track moves between sessions; lap times will not be comparable week to week until the duct arrives.
- **Cardboard walls have a life of weeks.** Budget an hour per session for the track, and plan on duct in the second half of the semester.
- **Out of scope for this budget.** The M1 lidar and any 1:5-scale vehicle, the Alameda track, a second car, and physical competitions. All are possible later; none are needed to get value from this car.

RoboRacer build guide and drawings: <https://f1tenth.readthedocs.io>

Team platform (simulator, devkit, manuals): <https://github.com/Pranman1/feb-racing>